HISTORY

It was in World War II (1940s) that this phenomenon was first noticed. Back then, it was cadmium that was used for electroplating, and there were many reports of failure of electrical hardware. Investigation into the failures found that some of the failures were being caused by what looked like tiny wires shorting out connectors or plates that weren't otherwise in contact. These "whiskers" of cadmium led to the problem being termed cadmium whiskers. Later, in 1948, Bell Telephone would report the same problems with their equipment. In an attempt to do away with cadmium whiskering, Bell changed to pure tin electroplating, but alas, the problem persisted. After a long period of studying the phenomenon, researchers were still left scratching their heads.

In the 1950s, researchers from US Steel did a study where they applied various forces to the substrate layer, and found that the more force applied to the base the faster the whiskers grew. With compressive forces of upto 7500 psi applied to the tin substrate, they found that the whisker growth was accelerated by as much as 10,000 times! Bell labs found that creating alloys of tin (Sn) that contained anywhere from 3 to 10 per cent lead (Pb) would significantly reduce or even do away with the whiskering problem.

In the 1970s the ESA (European Space Agency) made a blanket recommendation against the use of tin, zinc or cadmium in any spacecraft design. It gave allowance for the proven mitigation of whiskering using tin-lead combinations, however.

Until today, for tin-based electrical joints or circuits, adding lead is considered to be the only proper method of mitigating the formation of whiskers. Of course, lead poisoning is taken very seriously by pretty much any country on earth, and there are strict laws against the use of lead in consumer electronics. Pretty much every electronic circuit or device you have ever interacted with recently, does not contain any lead. All military equipment, however, has much stricter fault tolerance requirements, and isn't subjected to the same laws that consumer devices have to adhere to, and thus, contain the requisite amounts of lead in order to prevent whiskering. NASA and other space agencies also need to prevent against whiskering, and as a result have to pay extra to procure electrical and electronic devices that have lead in them, and of course, also have to take that much more trouble when disposing of older devices so as to not poison the environment with lead.

DAMAGE CAUSED

How serious can damage from whiskers be? Pretty darn serious, if you go by past events. Everything from aircraft radio communications in World War II to satellites dying suddenly, a lot has been attributed to these silly little metal hairs. In fact, it's not just satellites and radios, there are reports of heart pacemakers failing, and entire batches being recalled. Apart from the pacemakers in 1986, there were also apnea monitors failing because of zinc whisker formation in about 1994. Although this probably affects many more countries, there are documented failures of military technology because of whiskers. The failures include electronic systems in military airplanes, missiles, radar installations and more – of the stuff that's been declassified, that is. If all this wasn't scary enough, there have also been failures in power plants, and yes, some of them were nuclear power plants! From tripping warning systems, to even causing the shutting down of reactor cores, the little whiskers have played mayhem with power companies. In fact, high voltages seem to slightly speed up the whiskering process, and then there's always the additional stress and strain of heavy duty equipment, which we already know causes even more whiskering... Although highly disputed, a group at NASA studied failures in older Toyota Camry models (pre 2010), where accidents were being caused by the accelerator pedal seemingly having a mind of its own – causing sudden jumps in speed of 30 to 40 kmph, despite the driver never applying any more pressure to the pedal. Toyota and others who studied the problem dismiss whiskers as the cause, but the NASA scientists seemed sure of themselves, because whiskers were found when analysing the failed parts. Of course, the presence of whiskers doesn't necessarily mean that they are to blame for a fault, but one must consider them a possibility.

Doesn't that steel fastener look like it needs a shave?

But how do whiskers cause damage anway? Given that whiskers are usually less than a millimeter long and much thinner than a human hair, why do they cause so much damage?

Permanent short circuits: As the nomenclature suggests, this is often permanent damage, and causes equipment failure. All it takes to permanently damage something is for the circuit current to be below the fuse current of the whisker.

Temporary shorts: Unlike above, when the circuit current is higher than the fuse current of the whisker, it will result in a temporary short, then the whisker will fuse, and the circuit will return to normal. This may not sound like a big deal, but when it comes to precision circuitry, it can cause all sorts of problems. For one thing, it makes the circuit unreliable, and could damage sensitive components even with a temporary short. If the circuit is a data gathering circuit, a temporary short could result in false data being